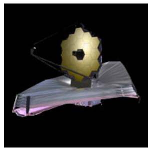

12264 - Cleaning Out the Archives: A Sweeping JWST Search for Dusty Supernovae

Cycle: 5, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Infrared (IR) observations of supernovae (SNe) at late times with the JWST are rapidly becoming a major focus of stellar evolution studies. Previous Spitzer results revealed that many SN types exhibit extended IR plateaus that persist for years to decades, yet the nature and heating source of the associated dust remain poorly understood. Determining the origin of these long-lived IR emissions could provide key insights into both the cosmic dust inventory and the mass-loss histories of evolved stars. JWST's unique combination of sensitivity and mid-infrared wavelength coverage now enables detection and characterization of the cooler dust components expected in diverse SN populations. Recent JWST observations have now uncovered unexpectedly large quantities of cold dust, renewing interest in dust-formation processes across different SN subclasses. Although several targeted JWST programs have begun to pursue this science, the limited number of individual objects restricts broader trend analysis over time and type. A complementary opportunity lies in the JWST archive: hundreds of serendipitous observations of nearby SNe (within 200 Mpc) will become

public at the start of Cycle 5, providing a rich, untapped resource. Our preliminary searches indicate that already multiple bright, dusty SNe have been captured in these datasets but remain unanalyzed. We therefore propose an Archival Research (AR) program to systematically examine these data for surviving dusty SNe or to establish upper limits. While the sample will be heterogeneous, it offers a cost-efficient and low-risk pathway to substantially expanding the JWST SN dust database and to guiding future monitoring campaigns.

OBSERVING DESCRIPTION

An AR proposal to search archival data at the positions of roughly 300 SNe for old, dusty SNe.